

Color and Colorimetry Multidisciplinary Contributions

Vol. XIX A

Edited by Filippo Cherubini and Andrea Siniscalco



www.gruppodelcolore.org

*Regular Member
AIC Association Internationale de la Couleur*

Color and Colorimetry. Multidisciplinary Contributions. Vol. XIX A

Edited by Filippo Cherubini and Andrea Siniscalco

Published by Gruppo del Colore - Associazione Italiana Colore

Research Culture and Science Books series (RCASB) ISSN: 2785-115X

ISBN 978-88-99513-23-8

DOI: 10.23738/RCASB.011

© Copyright 2024 by Gruppo del Colore – Associazione Italiana Colore

Piazza C. Caneva, 4

20154 Milano

C.F. 97619430156

P.IVA: 09003610962

www.gruppodelcolore.it

e-mail: segreteria@gruppodelcolore.org

Translation rights, electronic storage, reproduction
and total or partial adaptation with any means reserved for all countries.

Published in December 2024

Colour Meaning Matrix: A Tool for Connecting Concepts to Hue, Value and Saturation

Ingrid Calvo Ivanovic

Departamento de Diseño, Universidad de Chile

Contact: ingridcalvo@uchilefau.cl

Abstract

In the ideation phases of any design process, designers usually establish conceptual parallels between the creative ideas of the project and colour as a visual language, through colour meaning and associations. However, translating abstract or textual concepts to colour selection can be very challenging when dealing with complex or sophisticated meaning associations without physical or concrete visual referents to relate to. At the same time, hue has traditionally been considered the most salient descriptor of colour and colour–meaning associations, as much of what is commonly termed the "psychology of colour" has organized emotional responses to colour, mainly through a hue paradigm. In recent years, several studies have demonstrated that value and saturation have particularly strong meaning connotations and consistent effects on emotions. While models of colour–concept association based on hue alone may work for single-colour applications, they do not lend themselves well to design contexts where multiple colours are used simultaneously, such as environmental design. The paper presents the Colour Meaning Matrix (CMM), a practical tool to guide the selection of colours to represent conceptual associations based on semantic polar differentiation. The tool provides guidelines and suggestions to relate concepts to the scales of hue, but most importantly, lightness and saturation for the colour–concept association, allowing designers to select colours that convey the intended emotional and contextual meanings. The Colour Meaning Matrix has been tested in didactic contexts of higher education in design, with more than 300 students in two countries in Europe and Latin America.

Keywords: colour meaning, colour design, colour psychology, semantic differentiation

Introduction – Research background

Colour is one of the fundamental elements in design to convey meaning and emotions. Since objects inherently display colours, and colours evoke emotional responses that influence daily life, their semantic associations hold significant importance. A large number of studies point to connections between colour, meaning and emotion (e.g., Adams and Osgood, 1973; Byrnes, 1983; Kreitler and Kreitler, 1972; Schaie, 1962; Wexner, 1954). For instance, the impact of colour on emotions and preferences plays a crucial role in visual communication and marketing (Singh, 2006), and the colours of products can shape consumers' attitudes, either positively or negatively, and are instrumental in distinguishing products, brands, and corporate identities. The evocative and communicative nature of colour is not only related to personal intuition or experience. The chromatic aspect that characterises an image, a product or object or an environment can be articulated by the designer to stimulate perceptual and emotional responses in individuals, affecting their behaviour (Kotler, 1973). Colour can be used to visually communicate a specific mood. At the same time, colours can generate a psychological response in the observer: they may be identified as "mood colours," such as bright colours tend to cheer people, or neutral can be chosen to produce calm. Significance also influences our perception of colour and the function of designed objects; the meanings and use of colour may evolve from significant or cultural events and change based upon context (DeLong and Martinson

2012). Colours can represent a nation or institution, a group of people, a product, or an idea, and they can also symbolize and communicate social position.

- The hue paradigm: the traditional viewpoint

The emotional and semantic associations of colour have been commonly nominated with the name “colour psychology” or “colour meaning”, and its study for decades has mainly focused on the effects of hue (i.e., the attribute of a colour dependent on its dominant wavelength and independent of intensity or lightness, the most prominent quality by which we distinguish one colour from another) as the salient colour emotion descriptor. This approach has been described as the “*hue paradigm*”. People frequently associate certain emotions with particular hues (the subjective experience of anger is usually associated with the concrete concept of red because red correlates with the bodily experience (i.e., sensory, physiological) of increased blood flow during the experience of anger (Drummond and Quah, 2001). This approach has been largely studied: numerous research in the last century have investigated colour meaning only by inquiring on hue associations, without taking into account the other two attributes of colour, namely lightness and saturation. To approach the semantic aspects of colour taking into consideration only the aspects related to hue presupposes two significantly critical aspects: (i) the **plurality of meanings of a same hue**, and (ii) the **cultural-sensitive dichotomy**.

Firstly, the *plurality of meaning of the same hue* is pretty self-explanatory. Several research studies have found contradictory results (including antonymous concepts) for the meanings of the same hue. Moreover, some colours may be associated with several different emotions and some emotions are associated with more than one colour (Linton, 1999; Saito, 1996). When examining the meanings of red, for instance, this hue evoked a range of emotional associations—from negative connotations (such as danger) in achievement-related contexts to positive connotations (such as sexual readiness) in relational settings (Elliot and Niesta, 2008). Additionally, red was associated with both happy and angry faces (Palmer *et al.*, 2013). Furthermore, red was seen to be positive because it was associated with love and romance, while the negative aspects of red included having associations with fight and blood as well as Satan and evil (Naya and Epps, 2004). On other examples, blue was generally rated very positively (Adams & Osgood, 1973; Valdez and Mehrabian, 1994), yet it was linked to both calm and sad faces (Palmer *et al.*, 2013). Yellow was commonly paired with happy faces and upbeat music (*ibid.*), but when lightness and saturation were controlled, it was rated as the least pleasant colour in Valdez and Mehrabian's (1994) study. This plurality of colour meanings for the same hue has also been widely disseminated, including through literature, which has given rise to several misconceptions among the general public. But it raises more questions than it helps to answer: *How can the same hue be positive and negative at the same time or how can it simultaneously mean happiness and sadness, love and evil? What is its real meaning then? Do these pluralities of meaning help designers make colour decisions or do they confuse them?*

The other critical aspect of the hue paradigm is the *cultural-sensitive dichotomy*. This dichotomy proposes that on the one hand, in diverse cultures, different meanings are attributed to colour, according to the perceptions of value and local meanings of the territories or cultural affiliations. Therefore, colour conventions differ from one society to another. A well-known example is the two achromatic colours, black and white. Death and mourning are associated with the colour black in Western traditions, whereas in China the colour of death is white. Black can be associated not only with royalty, power, and wealth, but with death, mourning, and tragic events (Naya and Epps, 2004). Also in Asia, orange is a positive, spiritually enlightened, and life-affirming colour, whereas, in the United States, it is a colour of road hazards, traffic delays, and fast-food restaurants. However, different studies have concluded that the influence of cultural background on colour emotion and

meaning association is very limited (Gao and Xin, 2006). A good cross-culture consistency for the emotional connotation of colours was observed when comparing the results obtained in studies conducted in Sweden, Greece and Japan (*ibid*). Osgood studied anthropological field reports on five widely separated primitive cultures – Aztec and Pueblo Indian, Australian Bushman, Siberian Aborigine, Black (Uganda Protectorate) and Malayan – to obtain evidence on semantic parallelism. The generality of certain relationships was quite striking (Osgood *et al.*, 1967). Moreover, emotions were associated with colours similarly for US college students, Tzeltal-speaking people in Chiapas and Arawakan speaking Machiguenga in Peru. The dichotomy relies then on the fact that many (hue-based) colour–emotion associations vary between cultures while others appear consistently from one culture to the next. Both aspects of the dichotomy are contrasted when on the one hand, colour is undoubtedly, from an anthropological and ethnographic point of view, a powerful symbolic instrument that allows the differentiation and identification of cultures, but from a psychological perspective, there seems to be a human corporal and universal response to the stimulus of colours. So other questions arise: *Is colour meaning something agreed upon or is it inherent to human nature, such as a subconscious language? When designing colour, should we consider context-based meanings or universal, cross-cultural meanings?*

- More than hue: the contemporary approach

Due to the criticism that the “hue paradigm” has aroused as illustrated above, in recent years the study of the colour meaning and emotional associations has started to highlight the value and importance of the other two colour attributes: lightness (i.e. the light that a colour reflects, the attribute of colour consisting of a scale between black and white, how light or dark the colour is) and saturation (i.e. the intensity or purity of a hue, the extent of its colourfulness, and the strength or richness of a colour, indicating whether it is vivid or dull). Furthermore, different studies suggest that saturation and lightness are more important for the effects of colours on emotions than hue (Gao and Xin, 2006; Solli and Lenz, 2011). In 1994, Valdez and Mehrabian showed that changes in value (lightness) and chroma (saturation) can alter the emotional significance of a colour. In their study, they explored how participants responded to hue, lightness and saturation using the Pleasure-Arousal-Dominance (PAD) model. Their findings revealed that saturation had a significant and consistent impact on emotions being typically linked to Arousal, and lightness as associated with Dominance. Pleasure was connected to colours with both higher lightness and greater saturation (Valdez & Mehrabian, 1994).

Concerning the importance of lightness, Wright and Rainwater (1962) found that participants rated lighter colours more often using adjectives relating to happiness (happy, young, social, etc.). Valdez and Mehrabian (1994) showed also that people rated lighter chromatic and achromatic colours more positively. Similarly, Hemphill (1996) reported that viewing lighter colours elicited associations to positive affective states (e.g., white, pink, yellow, etc., associated with happiness and positivity) while viewing darker colours mainly produced associations to negative affective states (e.g. grey with sadness and boredom). Additionally, in a study by Adams and Osgood (1973), individuals from 23 cultures rated eight colour terms denoting higher lightness (white as compared to grey, as compared to black) as higher in pleasantness; while higher darkness (black) was the most consistently agreed upon colour concept as far as affect was concerned, finding it very bad, very strong and very passive (p. 147).

In addition to lightness, saturation might be more important in determining other affective associations, such as those with arousal or potency. Colours related to bodily expressions of positive (joy) emotion would have an overall higher lightness and saturation value than would colours chosen to go best with negative (fear) emotion (Dael *et al.*, 2016). Saturation has a more important role in determining the intensity of affective associations, such as those with arousal or potency, and it also

seems to have some consistent influence on average active/passive colour judgements. For Dael *et al.* (*ibid*), all three colour attributes are likely to be important when processing affective nonverbal person information, though not independently from each other. Related to this, Wright and Rainwater (1962) found that some meanings or associations hardly depend on hue, but in a combination of lightness and saturation of the specific colour. Some examples of this are the concept of “happiness”, which usually means an increase in lightness of a colour, and where saturation also makes a substantial contribution to it. Thus, the lighter or the more saturated the colour, the more “happiness” it connotes. Or the concept of “showiness”, which depends on lightness and saturation, but with a different emphasis: it is closely related to a high saturation. Interesting is the case of “forcefulness”, which can be affected by lightness and saturation distinctly: the darkest colour is, the strongest, but at the same time, a saturated colour is perceived as strong; high darkness and high saturation are not compatible in colour appearance: they can exist only apart from each other.

- Colour meaning decisions in the conceptual phase of design

In the ideation phases of any design process, designers usually establish conceptual parallels between the creative ideas of the project and colour as a visual language, through colour meaning associations. However, the prevailing model in much of the literature and resources available online on colour meaning for design is still the “hue paradigm”, which is not very useful for designers to tackle colour meaning challenges in real projects. In addition to the criticisms proposed above, Divers (2023) provides two other reasons for the need to abandon the “hue paradigm” in design practice and to embrace a perspective that considers the other two colour attributes: first, research findings have limited practical application because colours are usually studied individually, in isolation, to control the effects of colour interaction. However, designers are primarily tasked with managing the interplay of multiple colours, making this focus on single colours less relevant to their work; second, generalizations about hues typically refer to their full-saturation versions; for instance, red can be “exciting” in its most vivid form but tends to lose that effect when presented in muted, pale, or darker shades. Interestingly, designers tend to use muted colours more often than vibrant ones, which means research in this area often offers little practical guidance.

All of the above stresses the fact that designers **need practical tools and resources that may help them to make decisions related to colour meaning for real projects**. These tools must be updated and consider state-of-the-art knowledge, the contemporary approach to colour meaning beyond the “hue paradigm”. Some considerations to colour synaesthesia (i.e. the condition in which stimulation in one sensory or cognitive stream involuntarily, or automatically, leads to associated internal or external experiences in a second unstimulated sensory or cognitive system) can be helpful in the case of projects that may stimulate sensitive responses on users (such as the evocation through colour of the senses of taste, touch or smell in the case of some packaging projects, or the relation of colour with sound in projects related to music, among others). Another way to go is reviewing the contemporary aspects of colour associations concerning the final product's contextual, emotional, and cultural variables. These actions are usually supported by creating mood boards and working with design tools that may help translate abstract features of the creative concept into more concrete formal aspects or design elements.

Methodology

- The semantic differential method (SD)

Studies on colour emotion and meaning associations (several from the ones cited above) have usually examined colour meaning indirectly with the aid of psychological scales, such as the semantic

differential (SD) method, developed by Osgood et al. (Osgood *et al.*, 1967) for the analysis of meaning and is particularly useful to assess subjective judgments and attitudes. The SD is a psychological scaling method developed to understand how people perceive the meaning of words and concepts. Instead of asking individuals to describe the meaning of a word, they are asked to rate it on a series of bipolar adjective scales (e.g., *good-bad*, *strong-weak*, *active-passive*). The notion of using polar adjectives to define the termini of semantic dimensions grew out of research on synaesthesia during the first part of the XX century. Interrelationships among colour, mood, and musical experiences were studied, where subjects were asked first to relate music to a mood circle, and then, to select appropriate colours for those moods, resulting in consistent relations (*ibid*, pp.20-21). From those studies, the imagery found in synaesthesia was intimately tied up with language metaphor, and both of them represented semantic relations. The colour-music synaesthesia could then be described as the parallel alignment of two or more dimensions of experience, definable verbally by pairs of polar adjectives (*ibid*, p. 23).

When the SD method is used to assess colour meaning, usually, each bipolar scale is defined by a pair of polar (opposite-in-meaning) adjectives (e.g., joyful, sad, hard or soft), and the subject judges how a given colour is more related to one of the other pole of the scale. Each judgment reflects a choice and helps position the concept as a point within the semantic scale (from left to right). The more scales used, and the more representative those scales are, the more accurately this point reflects the operational meaning of the concept. The difference in meaning between two colours depends on the difference in how they are positioned within a shared scale (e.g., hot ___ X ___ cold). Osgood et al. (1967) also discovered that people's ratings consistently boiled down to *three core dimensions of meaning*, which he referred to as:

- a. Evaluation, whether something is good or bad, desirable or undesirable.
- b. Potency, how strong, powerful, or weak something is.
- c. Activity, whether something is active, dynamic, or passive.

In a nutshell, the overall results of applying SD scales for assessing colour demonstrate that the *evaluation dimension* is strongly associated with lightness, while *potency* tends to go with higher darkness and high saturation. *Activity* is most strongly associated with saturation, but also with brightness, where white is more active than both grey and black. The SD method usually demonstrates that colour emotion and meaning connotations are mainly connected to lightness and saturation, and less connected with hue (Adams and Osgood, 1973).

- Application of the SD method with design students

A set of exercises for the construction of semantic colour palettes, that is, colours chosen according to their meaning, were carried out with students of the “Colour Application” course of the third year of the Design program and the Arts and Crafts program at two Chilean universities. Additionally, exercises were carried out within the “Colour” teaching module with students in the first year of the Product Design program at an Italian university. All the exercises were carried out from 2021-2024, with more than 300 students.

Starting from a colour-music synaesthesia exercise, each student was given two pieces of popular instrumental music, with different moods (one joyful, other calm). The assignment was (*step 1*) to listen to the music and to write down concepts, words, feelings or ideas that were evoked by the musical mood for each piece. In parallel, (*step 2*) the students had to create 2 colour palettes, of ten colours each, that represented the mood of each music piece listened (one for each song). Subsequently, (*step 3*) students had to choose the most representative word or concept for each song

(resulting in 2 words in total) and create a new composed concept with both words (e.g. by using the words “movement” and “disconnection”, a student could create the concept of “*disconnected movement*”), and (step 4) compare both words to four given bipolar scales for semantic differentiation: the 3 original Osgood’s dimensions of meaning, namely, the *evaluation scale* (positive/negative), the *potency scale* (strong/weak), and the *activity scale* (active/passive); and a fourth novel scale proposed by the researcher, called the *understanding scale* (simple/complex). The SD was used for facilitating the students’ comprehension of the semantic mood of each concept and therefore, define their colour moods. Finally, with this judgement, (step 5) definitive 5-colours palettes were created, from the conceptual and colour moods defined, by using five colours from among the ones chosen in step 2. The application of this exercise after four years resulted in the creation of more than 1000 colour palettes (an average of 3,5 palettes per student), and around 350 composed concepts that were evaluated with the SD method. During this experience, collective discussions and different analyses were made throughout the years with the students, to understand what the colour appearance and therefore the attributes of the colours, were for each polarity. Some of the questions posed were: *How is the lightness of concepts with positive meanings? How is the saturation of colours related to active meanings? How are the colour attributes of a complex concept? What hues represent better the concept of strong or weak?* The answers to these and many other questions were obtained from the visual evidence, namely, the colour palettes created by the students and their comparison.

Results: The Colour Meaning Matrix (CMM)

From the collective analysis and discussion, the creation of the Colour Meaning Matrix (CMM) was proposed, as a practical tool to guide the selection of colours to represent conceptual associations based on semantic polar differentiation. The tool provides guidelines and suggestions to relate ideas

COLOUR MEANING MATRIX (CMM)						
	TYPE OF CONCEPT <i>how the concept or idea is perceived?</i>	OTHER SIMILAR COLOUR MOODS	HUE ORIENTATION <i>What hues?</i>	LIGHTNESS ORIENTATION <i>What lightness?</i>	SATURATION ORIENTATION <i>What saturation?</i>	
EVALUATION	Positive	good beautiful clean valuable kind pleasant happy	sacred nice honest fair rich clear healthy	Any. Basic hues that are easy to name and recognise (commonly known as primaries or secondaries). Also white, off-whites and pastels.	Medium to high. Usually intense colours, except when using white.	
	Negative	bad ugly dirty worthless cruel unpleasant sad	profane awful dishonest unfair poor hazy sick	Any. Very mixed hues (commonly known as tertiaries), shades which are more difficult to name.	Very low, close to zero. Dark colours. Low. Muted colours.	
POTENCY	Strong	strong large heavy thick hard deep	brave bass rough rugged wide	Any. Basic hues that are easy to name and recognise (commonly known as primaries or secondaries). Also black and dark shades.	Medium to low. Usually dark colours are perceived as strong.	Medium to high. Also saturated colours can be perceived as strong.
	Weak	small weak light weight thin soft shallow	cowardly treble smooth delicate narrow	Any. Preferably neutral shades, earthy tones, dirty hues, chromatic grays or achromatic colours.	Medium to high.	Low.
ACTIVITY	Active	active fast hot alive sharp	angular young ferocious tense	Any. Basic hues that are easy to name and recognise (commonly known as primaries or secondaries).	Medium.	High. Intense, vivid colours.
	Passive	passive slow cold dead dull	rounded old peaceful relaxed	Any. Preferably neutral shades, earthy tones, dirty hues, chromatic grays or achromatic colours.	High or low.	Low.
UNDERSTANDING	Simple	basic easy clear uncomplicated minimal straight direct usual	effortless uncomplicated elementary minimal straightforward homogeneous	Any. Basic hues that are easy to name and recognise (commonly known as primaries or secondaries). Also, achromatic colours.	Medium. Can be also high or low only when using achromatic colours.	High. Low when using achromatic colours.
	Complex	elaborate complicated intricate detailed layered unusual	challenging sophisticated advanced multi-faceted convoluted heterogeneous	Any. Very mixed hues (commonly known as tertiaries), shades which are more difficult to name.	High or low.	Low.

© Ingrid Calvo Ivanovic, 2024

© Ingrid Calvo Ivanovic, 2024

Figure 17: The Colour Meaning Matrix

to the scales of hue, but most importantly, lightness and saturation for the colour-concept association, allowing designers to select colours that convey the intended emotional and contextual meanings. The CMM is structured horizontally in eight rows: four dimensions of meaning with two poles each i.e., “evaluation” (positive and negative poles), “potency” (strong and weak poles), “activity” (active and passive poles), and “understanding” (simple and complex poles). Vertically, the CMM is structured in five columns, corresponding to: “type of concept” (how the concept or idea is usually perceived?), “other similar colour moods” (synonyms that can be used), “hue orientation” (what hues to select?), “lightness orientation” (what lightness to select?) and “saturation orientation” (what saturation to select?) (Fig. 1). The orientations included in the CMM are the result of the big amount of data collected, years of observation and the collective discussions indicated above.

The matrix is a tool for inquiring the semantic visual representation of the concepts or creative ideas, and it is meant to be used when the designer already has defined a concept for a project, by assessing it about the four different dimensions of meaning, e.g. starting from the first meaning dimension one could ask: *is my concept usually perceived as positive?*, if the answer to this question is *yes*, then the “positive” (row 1) orientations for hue, lightness and saturation may be followed as suggestions to choose colours with a that mood or character. If the answer to the same question instead is *no*, then it may mean that the concept is usually perceived as negative, and then the user may follow the orientations for the “negative” colour mood (row 2). But the answer to the question could be also *not applicable*, as not all ideas may be classified or judged as “positive” or “negative”, there are also some neutral concepts in terms of good or bad. If this is the case, the user of the CMM may proceed to evaluate her concept with the second meaning dimension, hence, “potency” and the second pair of polarities (rows 3 and 4) of strong and weak. Then, the logic of questions and answers is the same, until the concept or idea has been compared to all four meaning dimensions.

Conclusions

Since 2022, previous and partial versions of the CMM that is presented in this article were tested with students, which has led to the continuous improvement of the current tool. During the last three years, the CMM has been demonstrated to be useful for design students and professionals as it facilitates the process of colour decision-making, in terms of semantic associations. The CMM was not intended to be used as a formula, but as an orientation for the chromatic choices, and for this reason, it should not be conceived only as an answer but as a starting point for the semantic exploration of colour. The criteria and contents of the CMM are based on the evidence obtained from the literature review of the cited literature and others, on experimentation and four years of development of the tool from a large amount of analysed data, reflections and visual research through chromatic palettes. Future and ongoing developments consider the intersection of semantic dimensions and their polarities in a three-dimensional space, as well as an addendum or twin of the matrix created by incorporating synesthetic categories such as: wet/dry, tasty/insipid, rumorous/silent, among others.

References

- Adams, F.M. and Osgood, C.E. (1973) ‘A cross-cultural study of the affective meanings of colour’, *Journal of Cross-Cultural Psychology*, 4(2), pp. 135–156. doi:10.1177/002202217300400201.
- Byrnes, D.A. (1983) ‘Color associations of children’, *The Journal of Psychology*, 113(2), pp. 247–250. doi:10.1080/00223980.1983.9923582.
- Dael, N. et al. (2016) ‘Put on that colour, it fits your emotion: Colour appropriateness as a function of expressed emotion’, *Quarterly Journal of Experimental Psychology*, 69(8), pp. 1619–1630. doi:10.1080/17470218.2015.1090462.
- DeLong, M.R. and Martinson, B. (2014) *Color and design*. London: Berg Publications.

- Divers, E. (2023) 'The compass and the map: colour meaning and the colour design process', *Journal of the International Colour Association*, 32, pp. 44–51.
- Drummond, P.D. and Quah, S.H. (2001) 'The effect of expressing anger on cardiovascular reactivity and facial blood flow in Chinese and Caucasians', *Psychophysiology*, 38(2), pp. 190–196. doi:10.1017/s004857720199095x.
- Elliot, A.J. and Niesta, D. (2008) 'Romantic red: Red enhances men's attraction to women.', *Journal of Personality and Social Psychology*, 95(5), pp. 1150–1164.
- Gao, X. and Xin, J.H. (2006) 'Investigation of human's emotional responses on colors', *Color Research & Application*, 31(5), pp. 411–417. doi:10.1002/col.20246.
- Hemphill, M. (1996) 'A note on adults' color–emotion associations', *The Journal of Genetic Psychology*, 157(3), pp. 275–280. doi:10.1080/00221325.1996.9914865.
- Kotler, P. (1973) 'Atmospherics as a marketing tool', *Journal of Retailing*, 49(4), pp. 48–64.
- Kreitler, H. and Kreitler, S. (1972) *Psychology of arts*. Durham, NC: Duke University Press.
- Palmer, S. E., Schloss, K. B., Xu, Z. and Prado-León, L. R. (2013) Music-color associations are mediated by emotion. *Proceedings of the National Academy of Sciences of the United States of America*, 110(22), pp. 8836–8841. doi:10.1073/pnas.1212562110
- Saito, M. (1996) 'Comparative studies on color preference in Japan and other Asian regions, with special emphasis on the preference for white', *Color Research and Application*, 21(1), pp. 35–49.
- Schaie, K.W. (1962) 'A Q-sort study of color-mood association', *Journal of Projective Techniques*, 25(3), pp. 341–346. doi:10.1080/08853126.1961.10381048.
- Singh, S. (2006) 'Impact of color on marketing'. *Management Decision*, 44, pp. 783–789.
- Solli, M. and Lenz, R. (2011) 'Color emotions for multi-colored images', *Color Research & Application*, 36(3), pp. 210–221. doi:10.1002/col.20604.
- Linton, H. (1999) *Color in architecture: Design methods for buildings, interiors and urban spaces*. New York: McGraw Hill.
- Naya, K. and Epps, H.H. (2004) 'Relationship between color and emotion: a study of college students', *College Student Journal*, 38(3), pp. 396–405.
- Osgood, C.E., Suci, G.J. and Tannenbaum, P.H. (1967) *The measurement of meaning*. Chicago: University of Illinois.
- Valdez, P. and Mehrabian, A. (1994) 'Effects of color on emotions.', *Journal of Experimental Psychology: General*, 123(4), pp. 394–409. doi:10.1037//0096-3445.123.4.394.
- Wexner, L.B. (1954) 'The degree to which colors (hues) are associated with mood-tones.', *Journal of Applied Psychology*, 38(6), pp. 432–435. doi:10.1037/h0062181.
- Wright, B. and Rainwater, L. (1962) 'The meanings of color', *The Journal of General Psychology*, 67(1), pp. 89–99. doi:10.1080/00221309.1962.9711531.